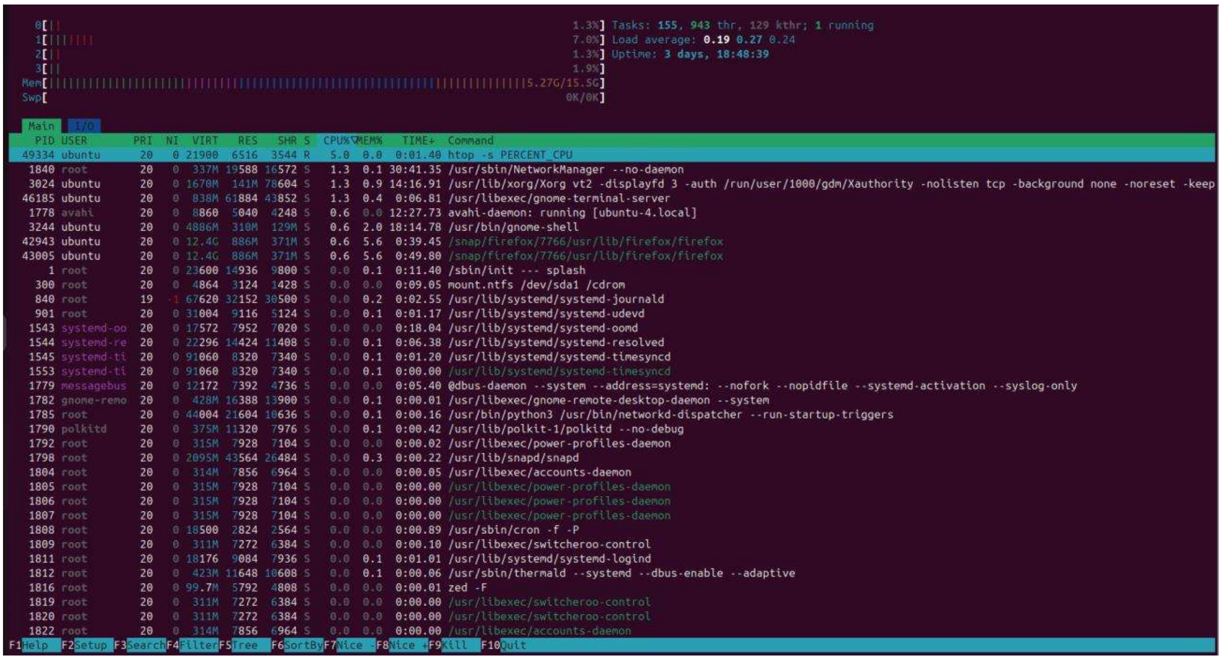
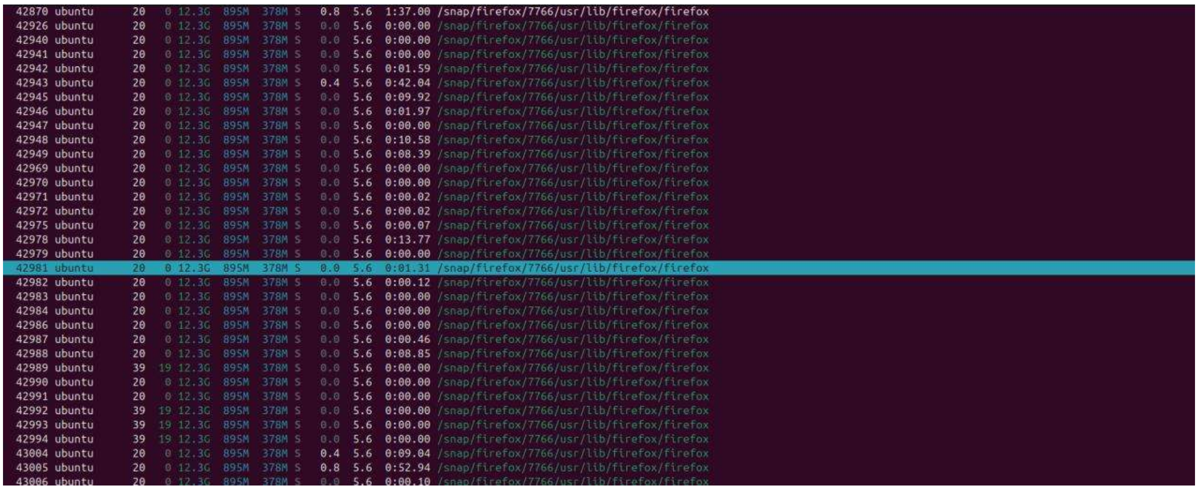
CWM WORK JOURNAL ASSIGNMENT 3

Q1:



The top consumer of CPU is the htop command itself. However, the list is continuously changing and sometimes it is one firefox command, which will occur when I start to type in this word document.

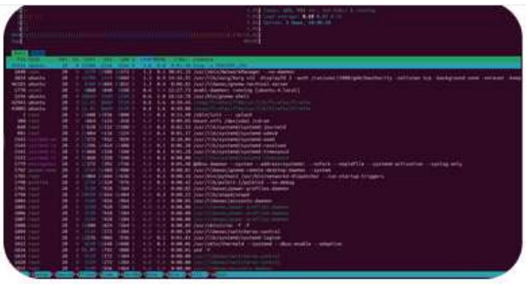
The top MEM consumer is a bunch of firefox threads, which are workers inside Firefox.



To be a top user in CPU and MEM, I can open some high-quality video games or play some movies on YouTube.

AI Use Q1

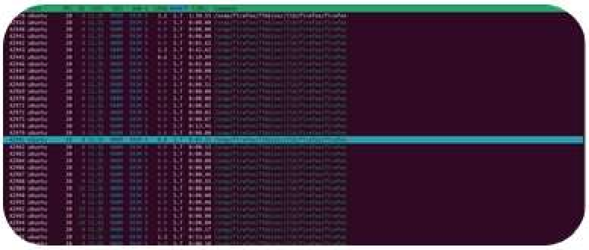
In this question, I asked GPT about the meaning of each column in this display, and what is the huge bunch of snap/firefox files?



What does each column mean?

This is **htop**, an interactive process monitor. Each row is a running process, and the columns describe its resource usage and status.

Column	Meaning
PID	Process ID. Unique number identifying the process.
USER	User who owns the process.
PRI	Scheduling priority. Lower values generally mean higher priority. Most normal processes run at 20.
NI	Nice value (-20 to 19). Lower = process gets more CPU preference. Higher = process is "nicer" and yields CPU more readily.
VIRT	Virtual memory size. Total memory the process can access, including code, libraries, mapped files, and swapped-out memory.
RES	Resident memory. Actual physical RAM currently used by the process.
SHR	Shared memory. Portion of RES that can be shared with other processes (e.g., shared libraries).
S	Process state.
CPU%	Percentage of CPU currently being used.



what are these snap commandes

Those are not separate Firefox programs. They are mostly **threads** belonging to the same Firefox process.

The clue is that they all have:

```
VIRT 12.3G
RES  908M
SHR  391M
Command /snap/firefox/.../firefox
```

and almost identical memory usage.

Why are there so many?

Modern browsers are multi-process and multi-threaded. Firefox creates:

- Main browser process
- Content processes (tabs/web pages)
- GPU process
- Network process
- Audio/video decoding processes
- Many worker threads

`htop` is currently configured to show **threads**, not just processes. Each thread gets its own PID/TID and appears as a separate row.

Q2:

For idle state, it means that the computer has little or no work to do. When the CPU is only running the htop instruction, with firefox opened at the background, the total CPU usage is about 13.1%, which indicates that the CPU has an idle period about 86.9% of all the time. The CPU usage means how much time it does work with proportion against the total operation time. Therefore, its compliment is the idle time, in which CPU does little work.

```
  Main  I/O
    PID USER       PRI  NI  VIRT   RES   SHR S  CPU%▽MEM%   TIME+  Command
  50004 ubuntu      20   0 21960  6668  3716 R   5.0  0.0  0:29.13 htop -s PERCENT_CPU
  42943 ubuntu      20   0 12.4G  911M  380M S   1.3  5.7  1:01.12 /snap/firefox/7766/usr/lib/firefox/firefox
  43079 ubuntu      20   0 12.4G  911M  380M S   1.3  5.7  1:26.39 /snap/firefox/7766/usr/lib/firefox/firefox
  50170 ubuntu      20   0 2512M 89484 72840 S   1.3  0.6  0:00.02 /snap/firefox/7766/usr/lib/firefox/firefox -contentproc -isForBrowser -prefsHandle 0:42730 -prefMapHandle 1:277145 -j
   1778 avahi       20   0  8860  5040  4248 S   0.6  0.0 12:31.18 avahi-daemon: running [ubuntu-4.local]
   1840 root        20   0  337M 19588 16572 S   0.6  0.1 30:52.10 /usr/sbin/NetworkManager --no-daemon
   3024 ubuntu      20   0 1671M  141M 78604 S   0.6  0.9 14:40.20 /usr/lib/xorg/Xorg vt2 -displayfd 3 -auth /run/user/1000/gdm/Xauthority -nolisten tcp -background none -noreset -keep
  42946 ubuntu      20   0 12.4G  911M  380M S   0.6  5.7  0:03.40 /snap/firefox/7766/usr/lib/firefox/firefox
  43015 ubuntu      20   0 12.4G  911M  380M S   0.6  5.7  0:37.94 /snap/firefox/7766/usr/lib/firefox/firefox
  46185 ubuntu      20   0  841M 65224 43852 S   0.6  0.4  0:15.64 /usr/libexec/gnome-terminal-server
  48736 ubuntu      20   0 3349M  731M  121M S   0.6  4.6  2:04.74 /snap/firefox/7766/usr/lib/firefox/firefox -contentproc -isForBrowser -prefsHandle 0:42730 -prefMapHandle 1:277145 -j
```

Q3:

Yes, the baseline does fluctuate overtime. This is because even when no work is assigned to CPU, the computer itself will reset timers or clear caches, which will lead to a sudden increment in total power consumption.

```
2.28
2.28
3.12
2.46
2.38
2.36
3.42
2.51
2.24
2.43
2.45
2.14
2.26
2.29
2.21
2.60
2.60
```

Q4a: Monitoring Baseline Operation

Busy%	Bzy_MHz	CoreTmp	PkgTmp	PkgWatt	CorWatt
1.85	1174	35	36	2.16	0.42
2.45	1271	35	37	2.22	0.48
2.11	1181	34	36	2.13	0.40
2.71	1636	35	36	2.40	0.66
2.03	1192	35	36	2.15	0.42
2.09	1214	35	37	2.17	0.44
1.72	1201	34	36	2.10	0.37
2.11	1178	34	36	2.19	0.45
2.42	1220	35	36	2.19	0.45
4.12	1658	35	36	2.40	0.66
3.47	1595	35	36	2.39	0.65
2.88	1487	35	37	2.28	0.54
16.82	3331	36	37	6.49	4.70
2.29	1417	35	37	2.24	0.50
2.60	1195	35	37	2.18	0.44
2.31	1281	34	37	2.20	0.46
1.75	1175	34	36	2.10	0.36
4.05	2163	35	36	2.71	0.96

Q4b: From this instruction, using the reported Busy%, if we take the complement of them, the results show that the idle time is much longer than previously estimated ones. This implies that the CPU could have different levels of idleness, where each of them represents a low operating power level.

Besides, another reasonable insight into this result is that the measurement of idleness of the computer will also depend on the instructions used for measurement. From previous results, the htop instruction itself is responsible for a significant CPU consumption when no other instructions are assigned. However, the sudo turbostat instruction used in this case might need much less CPU to run, thus showing a much higher idle time.

Q5:

CPU%c1	CPU%c3	CPU%c6	CPU%c7	Pkg%pc2	Pkg%pc3	Pkg%pc6	Pkg%pc7	Pkg%pc8
1.73	0.08	6.03	89.74	12.24	74.83	0.00	0.00	0.00
1.32	0.06	4.72	91.64	13.68	75.18	0.00	0.00	0.00
1.49	0.07	5.69	90.45	13.03	74.97	0.00	0.00	0.00
1.12	0.05	4.67	91.79	11.71	77.72	0.00	0.00	0.00
2.13	0.15	6.96	84.66	11.62	64.92	0.00	0.00	0.00
1.43	0.05	7.51	88.68	13.60	73.85	0.00	0.00	0.00
1.52	0.06	6.69	81.85	10.57	58.63	0.00	0.00	0.00
1.40	0.07	5.67	90.51	12.63	75.50	0.00	0.00	0.00
1.35	0.04	4.60	91.82	13.00	75.76	0.00	0.00	0.00
1.25	0.08	4.82	91.12	11.96	75.24	0.00	0.00	0.00
1.45	0.05	6.56	88.83	11.78	74.94	0.00	0.00	0.00
1.26	0.08	5.67	90.61	11.05	77.48	0.00	0.00	0.00
1.75	0.11	8.31	85.83	11.96	70.64	0.00	0.00	0.00
1.69	0.06	6.55	89.31	13.57	73.49	0.00	0.00	0.00
1.44	0.04	6.61	88.35	11.67	72.06	0.00	0.00	0.00
1.26	0.06	7.05	89.45	12.06	76.56	0.00	0.00	0.00
1.88	0.06	5.53	89.29	11.59	71.79	0.00	0.00	0.00

Q6:

A): When the stress-ng (matrix math) is assigned, the busy percentage in the power monitor bumped to 99.77%, but the CPU percentage drops to 0.23% for c1 and 0% for the rest.

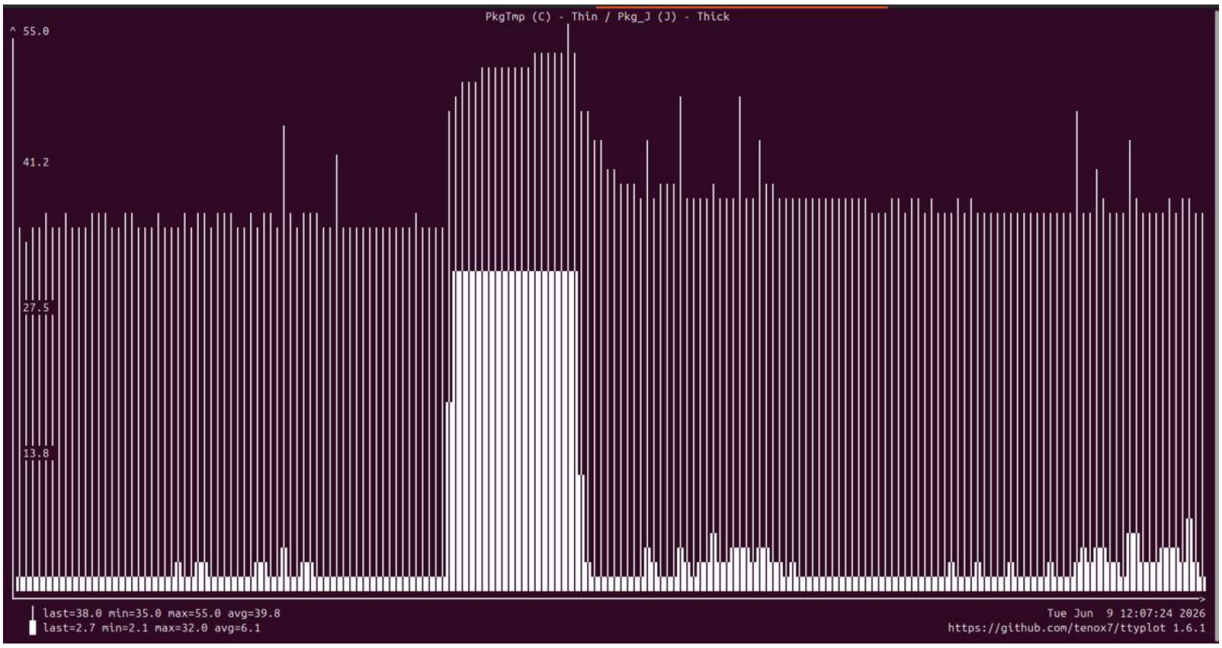
Results of experiment A:

2.88	1215	34	35	2.16	0.42
2.77	1196	34	36	2.13	0.39
5.80	1230	35	36	2.30	0.55
6.02	1524	41	41	2.52	0.78
20.17	2082	35	36	4.41	2.61
9.84	1290	35	36	2.50	0.74
16.59	1573	34	36	3.10	1.31
14.47	1355	36	36	2.81	1.04
94.60	3568	48	48	28.96	27.13
99.77	3600	49	49	31.26	29.44
99.77	3600	49	49	31.04	29.23
99.77	3600	50	50	31.09	29.27
99.77	3600	50	50	31.21	29.39
99.77	3600	50	50	31.29	29.47
99.77	3600	51	51	31.13	29.32
99.77	3600	51	51	31.26	29.47
99.77	3600	51	51	31.30	29.48
99.77	3600	52	52	31.44	29.62

3.07	0.34	9.06	77.10	9.63	57.43	0.00	0.00	0.00
3.27	0.53	6.88	44.75	6.01	27.63	0.00	0.00	0.00
0.24	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.23	0.00	0.01	0.00	0.00	0.00	0.00	0.00	0.00
0.28	0.01	0.02	0.09	0.00	0.00	0.00	0.00	0.00
0.23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
0.23	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2.62	0.37	6.29	17.48	3.06	8.67	0.00	0.00	0.00
5.91	1.02	11.58	56.23	7.55	33.85	0.00	0.00	0.00

B: Result of experiment B.

C: The baseline energy usage is quite small compared to the energy usage during stress testing. During stress testing, the CPU temperature jumps to 52 degrees and the power per Kg jumps to around 32, significantly large compared to idle time, when the CPU temperature is around 36 degrees and the power consumption is around 2.2. If the lab machines sit in the baseline state for most of the day, in this case, all power supplied to this computer is consumed during the idle time, which is just used to power this computer and doing nothing. However, if the computer is doing computations, it requires more power, but it is useful power supplying computer doing work.



Q7:

```
ubuntu@ubuntu:~/CWM-FDI/assignment1$ sudo turbostat -q --Joules --show Pkg_J python3 matmul_slow.py 500
n=500 reps=2 checksum=477458.000000
16.460132 sec
Pkg_J
176.67
176.67
ubuntu@ubuntu:~/CWM-FDI/assignment1$ sudo turbostat -q --Joules --show Pkg_J python3 matmul_fast.py 500
n=500 reps=2 checksum=477458.000000
11.516994 sec
Pkg_J
127.40
127.40
```

Yes, the fast code indeed consumes less power and has higher efficiency, and this matches my intuition. Fast leads to high efficiency as less cycles are run, and less instructions are performed in CPU for the fast algorithm. Therefore, reasonably, the faster algorithm should also consume less power.

Q8:

During the baseline period, no packages or cores indicating residency in deeper sleep states. There're only packages that indicate residency in 2 and 3, with major packages in 3. Drawbacks of a deeper sleep state might be crowded in other cores, in this case, for example, core 3. Besides, for deeper sleep states, their wakeup times are much longer compared to active states. Therefore, when a large number of requests arrives suddenly, the server or the computer might not be able to process these requests after the deeper sleep states cores wake up, indicating a huge delay for these requests. For example, video games or AI agents that will set up a huge number of requests suddenly might be affected by deeper sleep states.

CPU%c1	CPU%c3	CPU%c6	CPU%c7	Pkg%pc2	Pkg%pc3	Pkg%pc6	Pkg%pc7	Pkg%pc8
1.82	0.14	6.56	88.08	11.30	73.02	0.00	0.00	0.00
1.71	0.11	5.95	88.69	11.05	73.25	0.00	0.00	0.00
1.72	0.16	6.34	87.71	10.63	73.27	0.00	0.00	0.00
1.48	0.09	4.11	91.01	10.30	75.69	0.00	0.00	0.00
1.90	0.18	5.84	87.35	11.01	71.11	0.00	0.00	0.00
2.87	0.38	9.83	80.32	11.07	65.39	0.00	0.00	0.00
3.19	0.55	7.08	77.44	9.95	58.80	0.00	0.00	0.00
5.41	0.93	8.42	61.26	7.71	39.17	0.00	0.00	0.00
3.93	0.60	7.62	73.02	9.04	54.16	0.00	0.00	0.00
3.23	0.39	8.25	77.98	9.84	60.73	0.00	0.00	0.00
3.04	0.29	10.44	76.43	10.27	59.33	0.00	0.00	0.00
4.05	0.59	8.63	71.18	9.79	51.66	0.00	0.00	0.00
3.29	0.48	8.88	76.86	10.92	59.25	0.00	0.00	0.00
3.62	0.70	8.80	74.18	11.04	54.51	0.00	0.00	0.00
3.63	0.73	8.60	74.56	10.93	54.70	0.00	0.00	0.00
3.52	0.55	6.56	73.69	9.59	52.03	0.00	0.00	0.00
4.05	0.50	9.35	72.10	9.42	49.95	0.00	0.00	0.00
4.04	0.48	8.89	70.99	9.77	51.60	0.00	0.00	0.00
5.93	0.96	9.23	64.26	8.29	42.02	0.00	0.00	0.00
2.84	0.40	7.53	73.67	7.95	45.20	0.00	0.00	0.00
4.48	0.78	8.45	69.38	9.09	47.75	0.00	0.00	0.00
3.36	0.48	7.41	77.38	10.74	57.98	0.00	0.00	0.00
1.83	0.11	5.13	88.44	11.90	69.83	0.00	0.00	0.00
4.19	0.65	8.63	69.49	9.76	50.60	0.00	0.00	0.00
4.07	0.62	8.79	71.50	9.88	51.22	0.00	0.00	0.00
3.93	0.62	9.76	70.88	11.15	50.82	0.00	0.00	0.00
3.59	0.52	11.85	72.59	11.52	52.10	0.00	0.00	0.00
2.96	0.36	8.99	79.42	11.44	59.07	0.00	0.00	0.00
2.26	0.24	7.06	85.07	12.08	68.09	0.00	0.00	0.00
3.12	0.31	9.26	79.63	12.11	62.11	0.00	0.00	0.00
5.49	0.75	10.46	59.69	7.65	36.10	0.00	0.00	0.00
2.34	0.29	7.09	84.58	11.49	68.83	0.00	0.00	0.00
1.60	0.08	4.08	90.27	11.28	73.36	0.00	0.00	0.00
3.43	0.25	8.45	77.02	9.98	59.32	0.00	0.00	0.00
3.71	0.35	8.90	75.25	10.82	56.28	0.00	0.00	0.00
1.49	0.05	3.41	91.57	11.22	74.52	0.00	0.00	0.00
5.27	0.55	10.09	63.64	8.77	41.31	0.00	0.00	0.00
2.91	0.37	9.89	78.31	12.26	61.17	0.00	0.00	0.00
5.39	0.84	11.32	66.21	10.32	43.53	0.00	0.00	0.00
4.39	0.74	7.63	61.74	5.26	32.62	0.00	0.00	0.00

Q9:

Utilization of a data center will definitely affect its scale and computing power. For example, a cloud service provider for GPT or a training center for a LLM requires lots of CPUs and electric power to ensure enough computing power.

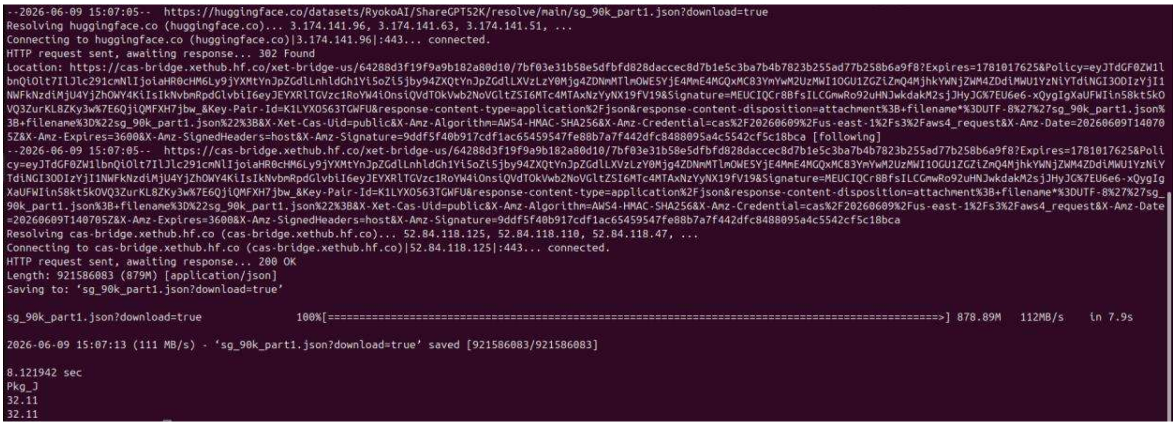
However, thermal conditions are major limitations to the design and scale of a data center, as CPUs need to operate below a certain temperature to achieve its maximum efficiency.

Therefore, submerged data centers and liquid cooling techniques are introduced to assist the CPUs to run at their maximum efficiency for a long time. Even though, thermal conditions will still limit the size and location of a data center as the effect of cooling techniques are still limited.

Q10: Network Activities

a. For a 1GB file, with transfer rate of 1Gb/s at full data rate, the transfer time is 1s.
b. When receiving at full rate, the active power is roughly 0.3 to 1 W. As the data is 1GB, the total energy consumed is from 0.3 to 1 J.
c. In this case, assuming all machines have the same download speed and the same power consumption, one machine needs to download the data file from the ethernet port and upload it to the next route. Therefore, the overall network energy impact should be the download and upload consumption times to the total number of machines. To make more accurate approximations, the idle power consumption of each machine is required as the machines need to be powered when waiting for the data. Besides, the transmission distance and time of the data file needs to be known to estimate the total idle energy.

Q11.



The theoretical result is far smaller than the experiment outcome. This is because the real download speed is much less than the theoretical value, which is 1GB, while the actual download speed is 112MB/s. Meanwhile, the theoretical value doesn’t include the energy used to power the computer. In this case, the program runs for about 8.1 seconds, which also means that a comparatively large part of the energy consumption of 32.11J is used to power the computer.

Q12:

-Data of streaming 2160p videos.

53.59	3359	47	47	15.38	13.39
37.06	2480	39	40	7.94	6.03
36.90	2047	39	40	6.02	4.12
36.18	2081	42	42	5.78	3.89
41.39	2642	42	42	8.96	7.03
84.48	3533	50	50	23.25	20.92
86.36	3600	51	50	24.66	22.24
74.74	3600	51	51	22.50	20.20
56.78	3600	48	48	18.16	16.01
73.96	3600	50	50	22.09	19.70
69.76	3600	51	51	21.41	19.08
50.90	3600	46	49	16.86	14.78
52.69	3600	52	52	17.37	15.25
48.70	3600	52	52	16.47	14.45
69.65	3600	52	52	21.92	19.74
61.35	3600	52	52	19.95	17.84
57.21	3600	51	51	18.97	16.83
66.24	3600	51	51	21.03	18.74
73.39	3600	45	45	22.28	19.94
36.63	2453	47	47	7.26	5.16
32.32	2421	48	48	6.64	4.64
74.56	3599	47	47	21.51	19.11
64.77	3599	53	53	19.89	17.68
68.91	3600	54	54	21.17	18.99
52.87	3600	53	53	17.64	15.56
51.62	3600	53	53	17.23	15.18
56.54	3600	52	52	18.40	16.27
49.89	3600	52	52	16.71	14.74
47.92	3600	52	52	16.25	14.26

-Data streaming 1080p videos.

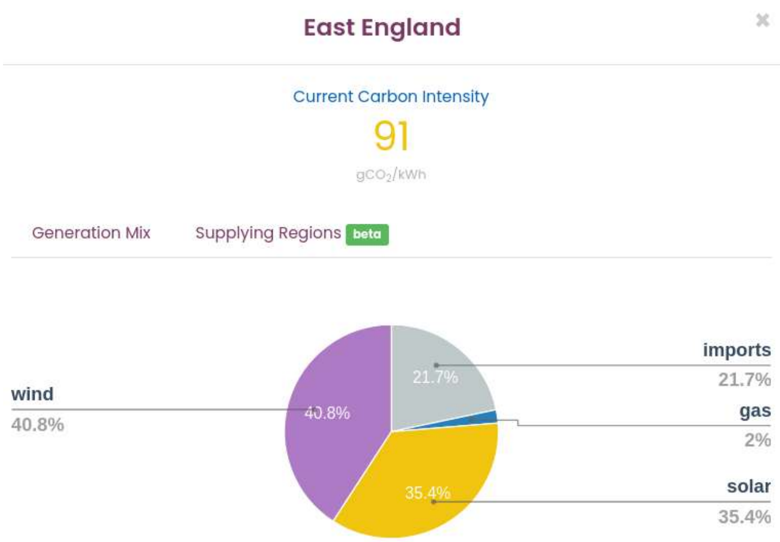
27.26	1307	40	42	3.17	1.32
44.45	2895	52	52	11.38	9.36
45.41	2981	42	43	12.12	10.09
34.76	1514	41	42	3.80	1.92
35.11	1563	41	41	4.16	2.27
31.58	1628	41	42	4.14	2.26
29.70	2245	41	42	5.77	3.88
35.85	2371	42	42	6.69	4.78
33.49	2294	46	46	6.06	4.16
33.92	2584	43	43	7.97	6.06
32.22	1536	42	42	3.88	2.00
28.57	1840	42	42	4.67	2.79
28.48	1759	43	43	4.48	2.61
34.62	1943	42	42	5.52	3.63
29.76	1745	42	42	4.57	2.70
27.02	1921	41	41	4.87	3.00
27.54	1808	47	46	4.57	2.70
27.69	1696	44	44	4.25	2.38
25.91	1804	40	42	4.36	2.50
26.87	1798	41	41	4.37	2.51
27.16	1755	40	41	4.39	2.52
27.25	1920	46	46	4.80	2.93
25.36	1921	41	42	4.76	2.89
28.18	1789	40	41	4.60	2.73
31.06	2134	43	43	5.93	4.04
38.90	2394	43	43	7.38	5.46
27.51	1934	41	42	5.09	3.21
29.19	1956	41	42	5.21	3.33

-Data streaming 480p videos.

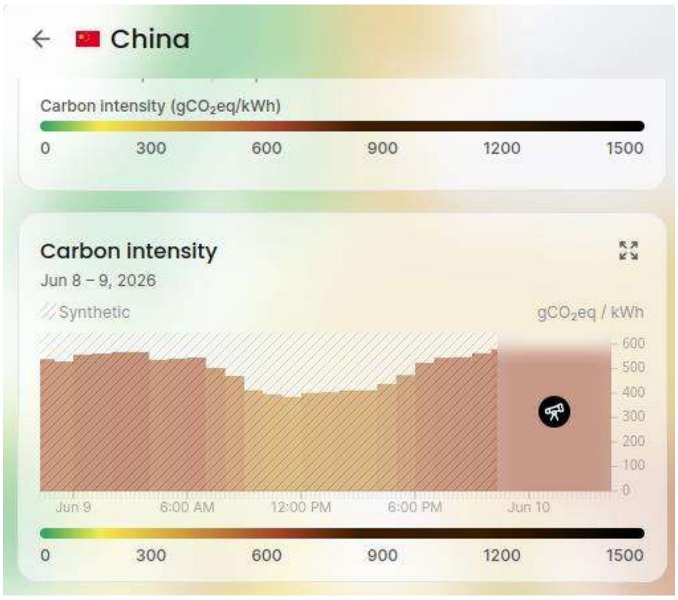
15.74	2687	40	41	5.03	3.24
26.40	1458	40	41	3.34	1.51
24.22	1331	40	41	3.04	1.23
24.97	1350	40	41	3.04	1.22
16.44	1295	39	41	2.69	0.90
16.01	1270	39	41	2.64	0.85
30.09	1655	48	48	3.89	2.06
14.50	1594	39	40	3.22	1.43
16.32	1468	39	40	3.02	1.24
17.82	1349	39	40	2.77	0.98
18.38	1307	39	40	2.76	0.97
22.05	1686	39	41	3.74	1.93
17.77	1321	39	41	2.71	0.92
18.10	1307	39	40	2.72	0.93
16.39	1314	39	40	2.70	0.91
15.95	1306	39	40	2.69	0.91
16.11	1315	39	40	2.72	0.93
16.83	1276	39	40	2.75	0.96
17.48	1288	39	40	2.74	0.95
27.39	1458	39	40	3.24	1.43
18.14	1262	39	40	2.75	0.96
23.57	1761	47	48	3.97	2.17
19.04	1907	39	41	4.17	2.34
17.96	1274	39	40	2.77	0.98

From the results, the higher the quality of video streaming, the more power consumption it has. The energy efficiency of video streaming gets higher when there's more power consumption as the proportion of energy used for videos against energy used for powering computers gets higher when there's more power consumption. Therefore, when the video streaming quality is higher, the corresponding energy efficiency is also higher.

Q13:



For east England, the current carbon intensity generated by electricity is 91g/kwh.



For China, on June 9th, 2026,

a. 3pm in the afternoon (UTC+1, which is 11pm at night for UTC+8), the carbon intensity is 580g/kwh.
b. 5am in the morning (UTC+1, which is 12pm at noon for UTC+8), the carbon intensity is 401g/kwh.

According to previous results, the energy required to run matmul_slow is 176.67J, and that required to run matmul_fast3 is 127.4J. To calculate carbon footprint, the energy consumption needs to be written in kwh and multiplied by the carbon intensity.

The carbon footprint of running these codes is:

Scenario	Carbon Intensity (gCO2e/kWh)	matmul_slow (g CO2e)	matmul_fast3 (g CO2e)	Carbon reduced
East England	91	0.00447	0.00322	0.00125
China (3 pm UTC+1)	580	0.02846	0.02053	0.00793
China (5 am UTC+1)	401	0.01968	0.01419	0.00549

The saving of carbon is quite different in each country, which is shown in the chart. Assuming one billion matrix operation in this scale is performed each day, the saving could be thousands of kilograms of carbon (1250kg, 7930kg, 5490kg), which is significant.

To minimize carbon emissions, it is definitely necessary to run my codes in countries with lower carbon intensity of electricity and run at the time when carbon intensity is the lowest in one day.

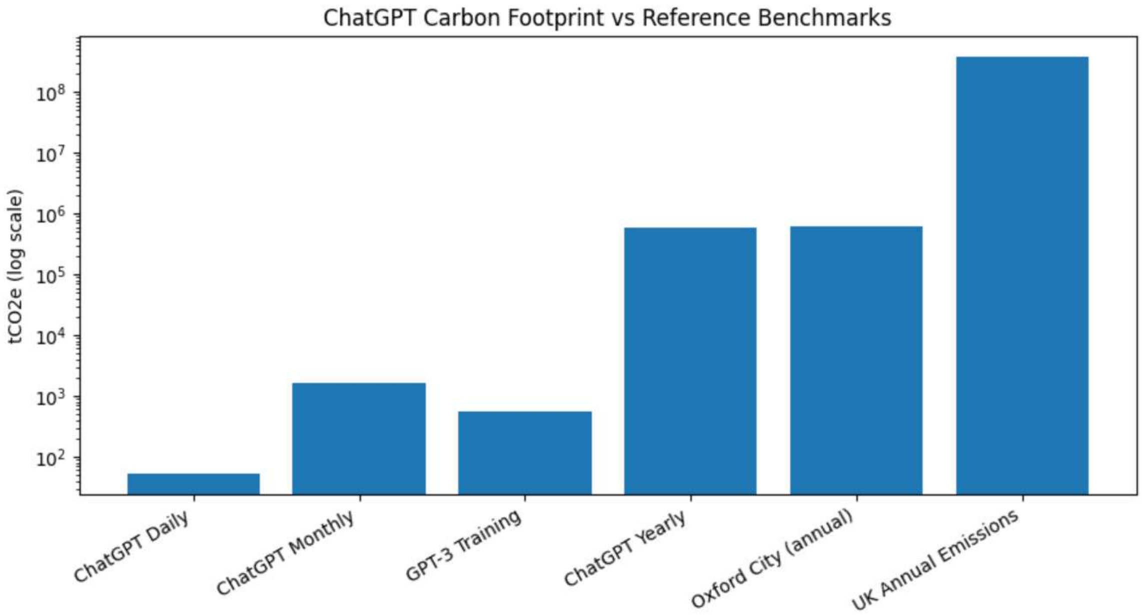
Q14:

For a standard text query, the energy consumption is 0.34Wh. Assuming 2.5 billion prompts per day is submitted to ChatGPT alone, using the carbon footprint in the U.S, the total carbon footprint of ChatGPT is shown:

-Carbon intensity: 63gCO2/kwh (U.S California, 7:45AM local time)

	Daily	Monthly	Yearly
Power per Queries(kwh)	0.00034	0.00034	0.00034
Number of Queries	2.5E+09	75000000000	2.7375E+13
Power Consumption	850000	25500000	9307500000
Carbon Footprint	53550000	1606500000	5.8637E+11

As a result, the carbon footprint of ChatGPT daily is 53550kg, which is gigantic.



The provided data suggest that ChatGPT's operational carbon footprint is substantially larger than the one-time carbon cost of training GPT-3. With an estimated 586,373 tCO2e per year, it is roughly 1,062 times greater than GPT-3 training emissions, comparable to the annual emissions of Oxford City, and equivalent to the emissions of approximately 146,000 UK homes. Astonishingly, to run ChatGPT, the carbon footprint produced is already in the same category as the carbon footprint, even larger than the carbon footprint. This simple result should raise more attention on reducing the carbon intensity in multiple developed countries and development of more efficient LLM models. [1234]

[1] GPT-3 training emissions (~552 tCO_2e).
[2] Oxford City emissions inventory and population statistics.
[3] UK Government, English Housing Survey 2022–2023 Energy Report.
[4] UK greenhouse gas emissions statistics (2024).

I'm using GPT here to help me gather carbon footprint data from other perspectives and assist me to generate a well-displayed graph comparing Carbon footprint of different sources.